Indian Stock Return Forecasting Using Machine Learning

UE24CS352A — Machine Learning Mini-Project | Team: [Member 1], [Member 2]

1. Problem statement

Can daily historical market data and technical features provide useful out-of-sample signals for forecasting the next five-trading-day return of an Indian-listed stock? This prototype compares regression models and examines a simple threshold-based long/flat strategy against buy-and-hold on the same held-out period. It is inspired by the Stanford CS229 report “Quantitative Trading with Machine Learning,” but is not a reproduction: it does not use the paper’s Volkswagen supply-chain dataset.

2. Dataset

The prototype supports daily OHLCV data from Yahoo Finance through yfinance, an uploaded CSV, and a bundled synthetic dataset for offline demonstration. Example symbols include RELIANCE.NS, TCS.NS, INFY.NS, and ^NSEI. Record the actual ticker, provider, dates, and row count after running the experiment. The bundled dataset is simulated and must never be reported as real market data.

3. Features

Features are computed from information available at or before each date: 1-, 5-, and 20-day returns; close-to-moving-average ratios; rolling return volatility; volume change; intraday range; and close location within the day's range. The target is the forward five-trading-day return, Close[t+5] / Close[t] − 1.

4. Methodology

We compare Elastic Net, a constrained Decision Tree, and Random Forest regressors. Observations are split chronologically into training, validation, and test subsets, with a five-row boundary embargo to reduce forward-target overlap. Model choice uses validation RMSE. The selected model is refit on training plus validation data and assessed on the untouched test set. Metrics are RMSE, MAE, and directional accuracy.

5. Implementation

trading_ml.py contains input validation, feature engineering, splitting, model fitting, metrics, data downloading, and backtesting. app.py provides a Streamlit dashboard for synthetic demo data, CSV upload, or Yahoo Finance download. The interface displays model metrics, feature importance, strategy curves, and downloadable results. A simple lagged long/flat signal with an adjustable threshold and transaction-cost assumption is compared against buy-and-hold.

6. Results and conclusion — COMPLETE AFTER RUNNING WITH REAL DATA

Ticker/source: [fill in] | Date range: [fill in] | Rows: [fill in] | Best validation model: [fill in]. Test RMSE: [fill in] | Test MAE: [fill in] | Direction accuracy: [fill in] | Strategy return: [fill in] | Buy-and-hold return: [fill in]. Do not submit invented results. Synthetic demo results are only a workflow check and cannot support claims about real-market performance.

7. Limitations and future work

Market data are noisy and non-stationary. Results may vary by ticker and date range. The strategy is simplified and does not fully model taxes, liquidity, market impact, or execution. Predictive association does not establish causation, and historical backtests do not guarantee future returns. Future work could add verified sector/index context and walk-forward testing. Confirm the course's assigned problem statement before final submission.